

Ex07. Introduction to Signals and systems

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1. Plot 2*2 Figures to visualize Laplace Transform

This assignment involves plotting the $x(t)$ graph derived from the given math problem, visualizing the s -plane of the Laplace transform using `abs()`, `real()`, and `imag()` functions, and identifying the coordinates of poles and zeros. For the numerical computation, time t is sampled at intervals of 0.01 from -5 to 5. Similarly, σ and the angular frequency ω both range from -5 to 5 with an increment of 0.05.

```
t = -5:0.01:5;
sigma = -5:0.05:5;
omega = -5:0.05:5;

[SIGMA, OMEGA] = meshgrid(sigma, omega);
s = SIGMA + 1j*OMEGA;

num = -5;
den = [1 1 -6];

x = @(t)exp(-3*t).*(t >= 0) + exp(2*t).*(t <= 0);
X = @(s) -5 ./ ((s + 3) .* (s - 2));

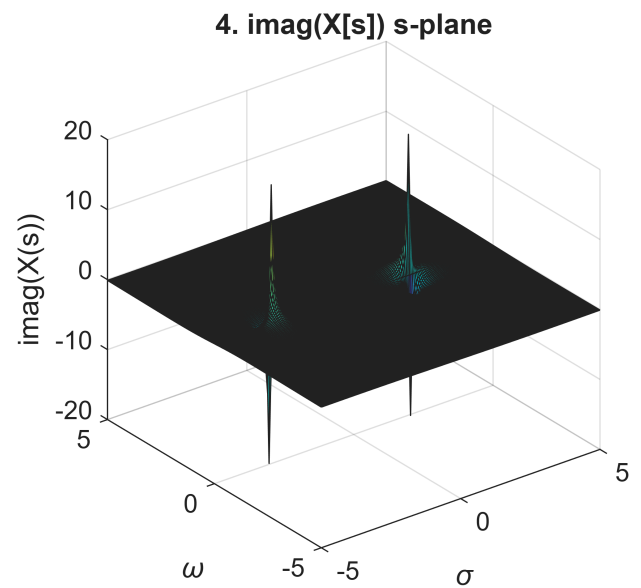
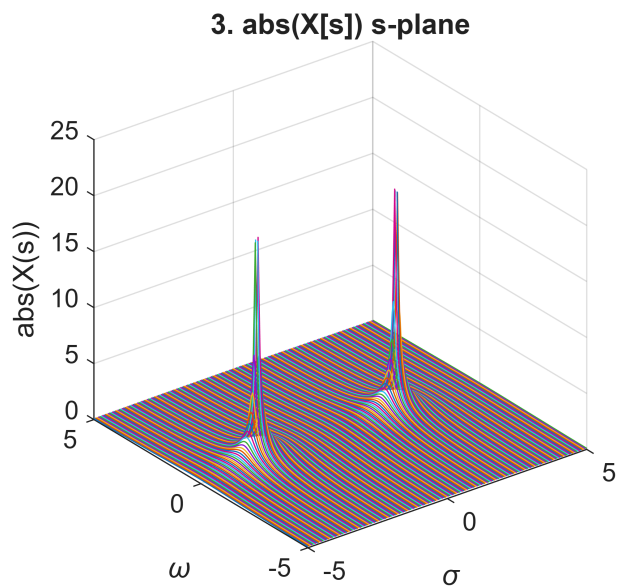
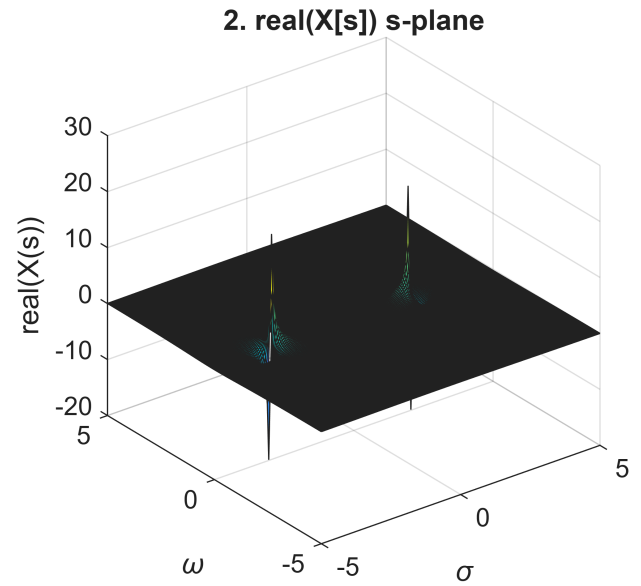
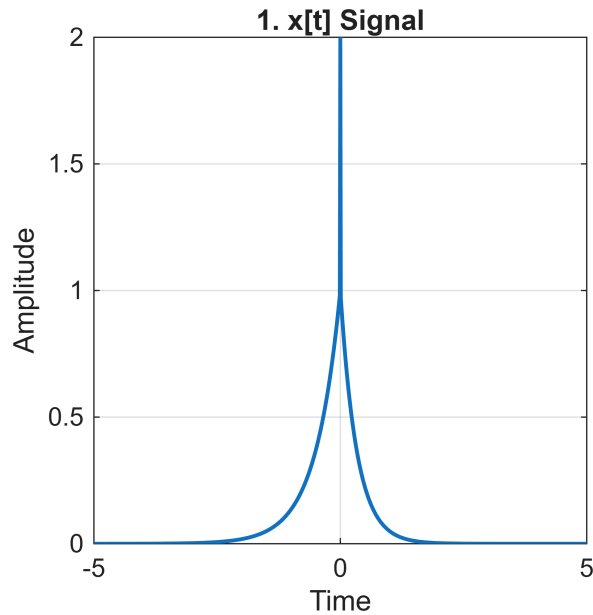
figure('Position', [100, 100, 800, 800]);

subplot(2, 2, 1);
p1 = plot(t, x(t), 'Linewidth', 1.5);
title('1. x[t] Signal');
xlabel("Time");
ylabel("Amplitude");
grid on;
subplot(2, 2, 2);
p2 = surf(SIGMA, OMEGA, real(X(s)));
title('2. real(X[s]) s-plane');
xlabel('\sigma');
ylabel('\omega');
zlabel('real(X(s))');
grid on;
subplot(2, 2, 3);
p3 = plot3(SIGMA, OMEGA, abs(X(s)));
title('3. abs(X[s]) s-plane');
xlabel('\sigma');
ylabel('\omega');
zlabel('abs(X(s))');
grid on;
subplot(2, 2, 4);
p4 = surf(SIGMA, OMEGA, imag(X(s)));
```

```

title('4. imag(X[s]) s-plane');
xlabel('\sigma');
ylabel('\omega');
zlabel('imag(X(s))');
grid on;

```



```

poles_coords = roots(den);
zeros_coords = roots(num);

```

```
disp('--- Task 5: Coordinates of poles and zeros ---');
```

```
--- Task 5: Coordinates of poles and zeros ---
```

```
disp('Poles on s-plane:');
```

```
Poles on s-plane:
```

```
disp(poles_coords);
```

```
-3  
2
```

```
disp('Zeros on s-plane:');
```

```
Zeros on s-plane:
```

```
if isempty(zeros_coords)  
    disp('None');  
else  
    disp(zeros_coords);  
end
```

```
None
```

```
%Since the numerator is a nonzero constant, there are no finite zeros
```

2. Math Problem

Below is the snapshot of my answer in fill-blank math problem.

$$\frac{1}{s+3} \quad \text{---} \quad \frac{-1}{s-2} \quad \text{---} \quad \frac{-1}{s-2} \quad \text{---} \quad \frac{-1}{s-2} \quad \text{---} \quad \frac{-1}{s-2}$$

$$\frac{1}{s+3} - \frac{1}{s-2} = \frac{-5}{(s+3)(s-2)} \quad \text{---} \quad \frac{-5}{(s+3)(s-2)} \quad \text{---} \quad \frac{-5}{(s+3)(s-2)}$$

$$s = -3, 2 \quad \text{---} \quad \text{None} \quad \text{---} \quad \text{None}$$